Fifth_

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```
library(deSolve)
library(ggplot2)
library(patchwork)
library(tidyr)
```

```
# Parameters

params_fixed <- list(

  # ART initiation
  gammaF = 0.85,
  gammaM = 0.89,

  # Natural mortality
  muF = 0.02,
  muM = 0.02,

  # HIV related mortality for untreated FSW and MC
  thetaF = 0.1,
  thetaM = 0.1,

  # Mortality among those on ART
  thetaTF = 0.01,
  thetaTM = 0.01,

  # Transmission probabilities
  betaMF = 0.0008,   # male -> female
  betaFM = 0.0004,   # female -> male

  # Contact rates
  cF = 1550,
  cM = 500,

  # Relative infectiousness on ART
  fT = 0.03,

  # Condom intervention
  condom_eff = 0.76,
  condom_covF = 0.50,
  condom_covM = 0.6,

  # PrEP intervention
```

```
  PrEP_eff = 0.53,
  PrEP_covF = 0.32,
  PrEP_covM = 0.60,

  # Circumcision
  circ_eff = 0.55,
  circ_cov = 0.50
)
```

```
# Model function
model <- function(t, state, p){

  with(as.list(c(state, p)),{

    # TOTAL POPULATIONS

    NF <- SF + IF + TF

    NM <- SM + IM + TM

    # RECRUITMENT

    recruitF      <- muF * NF
    recruitM      <- muM * NM

    # INFECTIOUS PREVALENCE

    prev_M <- (IM + fT * TM) / NM

    prev_F <- (IF + fT * TF) / NF

    # INTERVENTION REDUCTIONS

    red_M <- (1 - condom_eff * condom_covM) *
             (1 - PrEP_eff   * PrEP_covM)

    red_F <- (1 - condom_eff * condom_covF) *
             (1 - PrEP_eff   * PrEP_covF) *
             (1 - circ_eff   * circ_cov)

   # FORCE OF INFECTION

    lambdaF <- cF *
                    betaMF *
                    red_M *
                    prev_M

    lambdaM <- cM *
               betaFM *
               red_F *
               prev_F

    # FSW HIGH RISK
```

```
    dSF <- recruitF -
                lambdaF * SF -
                muF * SF

    dIF <- lambdaF * SF -
                (gammaF + muF + thetaF) * IF

    dTF <- gammaF * IF -
             (muF + thetaTF) * TF

    # MALE CLIENTS

        dSM <- recruitM -
           lambdaM * SM -
           muM * SM

    dIM <- lambdaM * SM -
           (gammaM + muM + thetaM) * IM

    dTM <- gammaM * IM -
       (muM + thetaTM) * TM

    # INCIDENCE

    incF <- lambdaF * SF

    incM <- lambdaM * SM

    list(c(dSF, dIF, dTF,
           dSM, dIM, dTM),

         incF = incF,
         incM = incM)
  })
}
```

```
# Initial conditions
state <- c(

  SF = 0.62,
  IF = 0.06,
  TF = 0.32,

  SM = 0.82,
  IM = 0.02,
  TM = 0.16
)
# Simulation time
times <- seq(0, 10, by = 0.1)
# CALIBRATION TARGETS
FSW_lower <- 0.30
FSW_upper <- 0.45
MC_lower <- 0.10
```

```r
MC_upper <- 0.25
```

```r
# Run model
run_model <- function(p){

  as.data.frame(
    ode(
      y = state,
      times = times,
      func = model,
      parms = p
    )
  )
}
# BASIC REPRODUCTION NUMBER (R0) with no interventions
# No condoms, no PrEP, no circumcision, no ART

calc_R0 <- function(p){

  p0 <- p

  # Remove all interventions
  p0$condom_covF <- 0
  p0$condom_covM <- 0

  p0$PrEP_covF <- 0
  p0$PrEP_covM <- 0

  p0$circ_cov <- 0

  p0$thetaF <- 0
  p0$thetaM <- 0

  DF <- 1/(p0$gammaF + p0$muF)
  DM <- 1/(p0$gammaM + p0$muM)

  K_MF <- p0$cF * p0$betaMF * DF

  K_FM <- p0$cM * p0$betaFM * DM

  R0 <- sqrt(K_MF * K_FM)

  return(R0)
}

# EFFECTIVE REPRODUCTION NUMBER (Re)
calc_Re <- function(p){

  DF <- 1/(p$gammaF + p$muF + p$thetaF)
  DM <- 1/(p$gammaM + p$muM + p$thetaM)

  red_M <- (1 - p$condom_eff * p$condom_covM) *
           (1 - p$PrEP_eff * p$PrEP_covM)
```

```
  red_F <- (1 - p$condom_eff * p$condom_covF) *
           (1 - p$PrEP_eff * p$PrEP_covF) *
           (1 - p$circ_eff * p$circ_cov)

  K_MF <- p$cF *
          p$betaMF *
          red_M *
          DF

  K_FM <- p$cM *
          p$betaFM *
          red_F *
          DM

  Re <- sqrt(K_MF * K_FM)

  return(Re)
}
```

```
# Objective function
# CALIBRATION OBJECTIVE FUNCTION
objective <- function(par){

  p <- params_fixed

  # Parameters to calibrate
  p$betaMF  <- par[1]
  p$betaFM  <- par[2]
  p$cF <- par[3]
  p$cM <- par[4]

  # Run model
  out <- run_model(p)

  # FSW prevalence
  FSW_prev <- tail(

    (out$IF + out$TF) /

    (out$SF + out$IF + out$TF),

    1
  )
   # Male client prevalence
  MC_prev <- tail(

    (out$IM + out$TM) /

    (out$SM + out$IM + out$TM),

    1
  )
```

```
  # Range-based FSW error
  if(FSW_prev < FSW_lower){

  err_FSW <- (FSW_prev - FSW_lower)^2

} else if(FSW_prev > FSW_upper){

  err_FSW <- (FSW_prev - FSW_upper)^2

} else {

  err_FSW <- 0

}

  # Range-based male client error
  if(MC_prev < MC_lower){

    err_MC <- (MC_prev - MC_lower)^2

  } else if(MC_prev > MC_upper){

    err_MC <- (MC_prev - MC_upper)^2

  } else {

    err_MC <- 0

  }

R0_now <- calc_R0(p)
err_R0 <- 0

if(R0_now < 1.2){

  err_R0 <- (1.2 - R0_now)^2

}

# Total objective
  error <- 100 * err_FSW +
           200 * err_MC  +
           50  * err_R0
  return(error)
}
# PARAMETER CALIBRATION

fit <- optim(

  par = c(
    betaMF  = 0.003,
    betaFM  = 0.001,
    cF = 1000,
```

```
    cM = 300
  ),

  fn = objective,

  method = "L-BFGS-B",

  lower = c(
    0.001,
    0.0003,
    800,
    20
  ),

  upper = c(
    0.01,
    0.005,
    4000,
    1200
  )
)

# CALIBRATED PARAMETERS

params_calibrated <- params_fixed

params_calibrated$betaMF  <- fit$par[1]
params_calibrated$betaFM  <- fit$par[2]
params_calibrated$cF      <- fit$par[3]
params_calibrated$cM      <- fit$par[4]

# Run calibrated model
out <- run_model(params_calibrated)
```

```
prev_df <- data.frame(

  Time = out$time,

  FSW =
    (out$IF + out$TF) /

    (out$SF + out$IF + out$TF),

  MC =
    (out$IM + out$TM) /
    (out$SM + out$IM + out$TM)
)
FSW_prev <- tail(prev_df$FSW, 1)
MC_prev  <- tail(prev_df$MC, 1)

cat("Final FSW prevalence =", round(FSW_prev, 3), "\n")
```

```
## Final FSW prevalence = 0.422
```

```r
cat("Final MC prevalence  =", round(MC_prev, 3), "\n")
```

```
## Final MC prevalence  = 0.19
```

```r
if(MC_prev >= MC_lower &&
   MC_prev <= MC_upper){

  cat("Calibration target achieved\n")

} else {

  cat("Outside calibration range\n")

}
```

```
## Calibration target achieved
```

```r
cat("Basic reproduction number (R0) =",
    round(calc_R0(params_calibrated), 2), "\n")
```

```
## Basic reproduction number (R0) = 2.14
```

```r
cat("Effective reproduction number (Re) =",
    round(calc_Re(params_calibrated), 2), "\n")
```

```
## Effective reproduction number (Re) = 0.72
```

```r
cat("\nCALIBRATED PARAMETERS\n")
```

```
##
## CALIBRATED PARAMETERS
```

```r
print(params_calibrated[c(
  "betaMF",
  "betaFM",
  "cF",
  "cM"
)])
```

```
## $betaMF
##      betaMF
## 0.005274918
##
## $betaFM
##      betaFM
## 0.002299953
##
## $cF
##       cF
```

```
## 1000.002
##
## $cM
##       cM
## 300.0077
```

```
# Prevalence plots
p1 <- ggplot(prev_df,
             aes(Time, FSW)) +

  geom_line(linewidth = 1.2) +

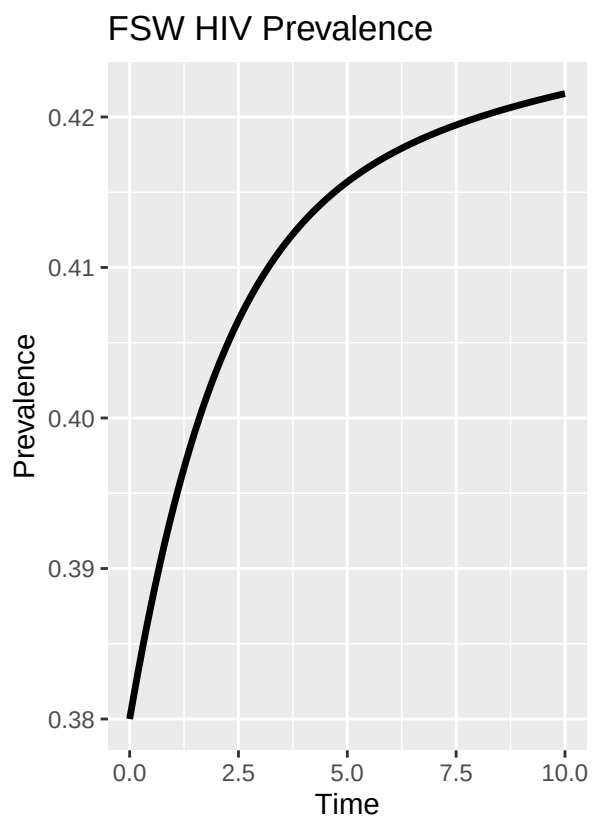
  labs(
    title = "FSW HIV Prevalence",
    y = "Prevalence"
  )

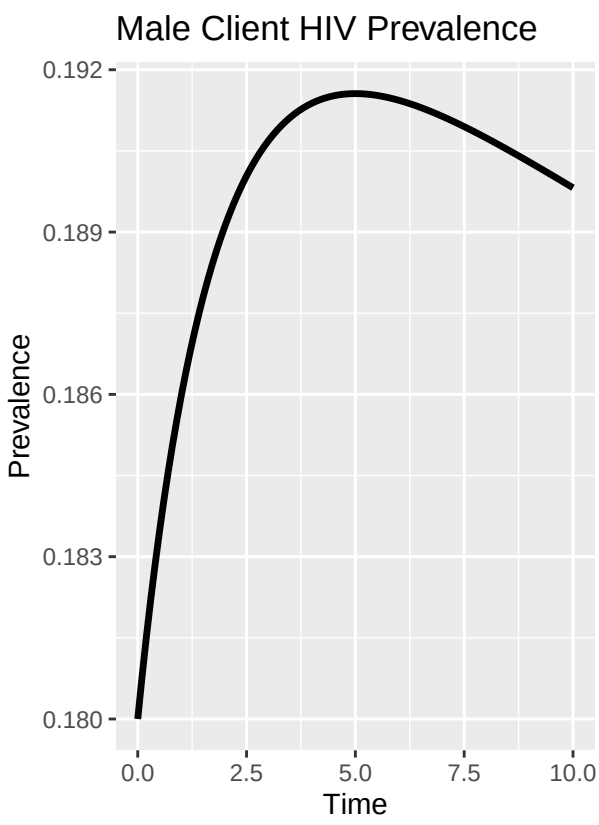
p2 <- ggplot(prev_df,
             aes(Time, MC)) +

  geom_line(linewidth = 1.2) +

  labs(
    title = "Male Client HIV Prevalence",
    y = "Prevalence"
  )

p1 + p2
```






```
# Incidence
inc_df <- data.frame(
  Time = out$time,
  FSW = out$incF,
  MC = out$incM
)

g1 <- ggplot(inc_df,
             aes(Time, FSW)) +

  geom_line(linewidth = 1.2) +

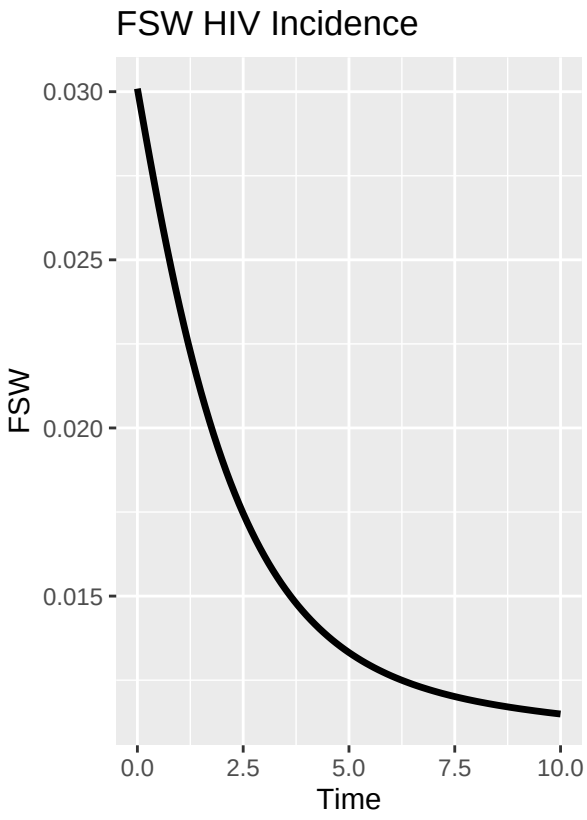
  labs(
    title = "FSW HIV Incidence"
  )

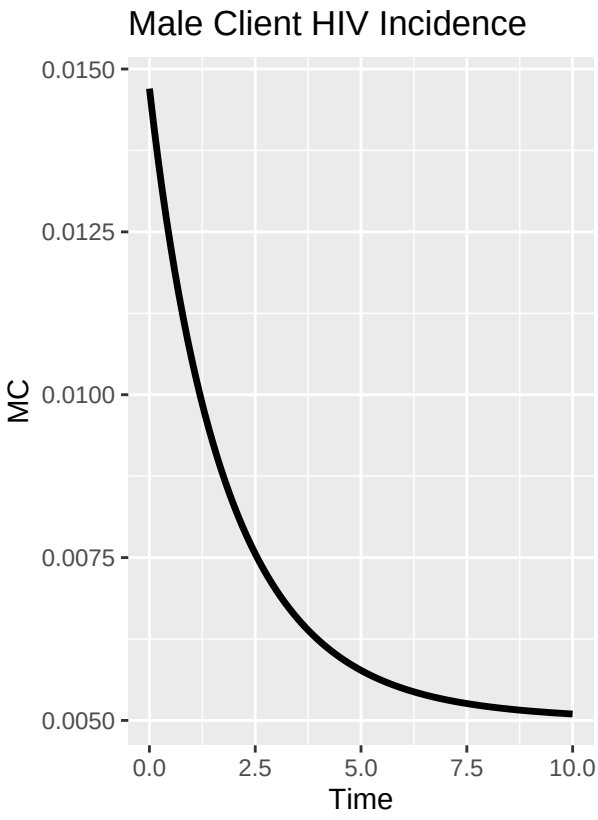
g2 <- ggplot(inc_df,
             aes(Time, MC)) +

  geom_line(linewidth = 1.2) +

  labs(
    title = "Male Client HIV Incidence"
  )

g1 + g2
```






```r
scenarios <- list(

  # BASELINE
  Baseline = params_calibrated,

  # NO INTERVENTION
  No_Intervention = within(params_calibrated, {

    condom_covF <- 0
    condom_covM <- 0
    PrEP_covF <- 0
    PrEP_covM <- 0
    circ_cov <- 0
  }),

  # SINGLE INTERVENTIONS
  Condoms_Only = within(params_calibrated, {

    condom_covF <- 0.85
    condom_covM <- 0.85
  }),

  PrEP_Only = within(params_calibrated, {

    PrEP_covF <- 0.90
    PrEP_covM <- 0.90
```

```
  }),

  Circumcision_Only = within(params_calibrated, {

    circ_cov <- 0.90
  }),

  # TWO-COMBINATION PACKAGES
  Condoms_PrEP = within(params_calibrated, {

    condom_covF <- 0.85
    condom_covM <- 0.85

    PrEP_covF <- 0.90
    PrEP_covM <- 0.90
  }),

  Condoms_Circ = within(params_calibrated, {

    condom_covF <- 0.85
    condom_covM <- 0.85

    circ_cov <- 0.90
  }),

  PrEP_Circ = within(params_calibrated, {

    PrEP_covF <- 0.90
    PrEP_covM <- 0.90

    circ_cov <- 0.90
  }),

  # FULL COMBINATION
  All_Interventions = within(params_calibrated, {

    condom_covF <- 0.85
    condom_covM <- 0.85

    PrEP_covF <- 0.90
    PrEP_covM <- 0.90

    circ_cov <- 0.90
  })
)
# Scenario outputs
outputs <- lapply(scenarios, run_model)
```

```
summary_table <- do.call(
  rbind,

  lapply(names(outputs), function(n){
```

```
    out <- outputs[[n]]
    p   <- scenarios[[n]]

    FSW_prev <- tail(
      (out$IF + out$TF) /

      (out$SF  + out$IF  + out$TF),
      1
    )

    MC_prev <- tail(
      (out$IM + out$TM) /
      (out$SM + out$IM + out$TM),
      1
    )

    data.frame(
      Scenario = n,
      FSW_prev = round(FSW_prev, 3),
      MC_prev  = round(MC_prev, 3),
      R0 = round(calc_R0(p), 2),
      Re = round(calc_Re(p), 2),
      FSW_cum_inc = round(sum(out$incF), 2),
      MC_cum_inc  = round(sum(out$incM), 2)
    )
  })
)
summary_table
```

```
##              Scenario FSW_prev MC_prev   R0   Re FSW_cum_inc MC_cum_inc
## cF           Baseline    0.422   0.190 2.14 0.72        1.57       0.69
## cF1   No_Intervention    0.890   0.483 2.14 1.93        6.67       4.17
## cF2      Condoms_Only    0.352   0.158 2.14 0.44        0.77       0.30
## cF3         PrEP_Only    0.367   0.162 2.14 0.50        0.95       0.35
## cF4 Circumcision_Only    0.399   0.170 2.14 0.60        1.31       0.45
## cF5      Condoms_PrEP    0.331   0.147 2.14 0.30        0.52       0.17
## cF6      Condoms_Circ    0.346   0.150 2.14 0.36        0.69       0.20
## cF7         PrEP_Circ    0.358   0.153 2.14 0.42        0.84       0.24
## cF8 All_Interventions    0.328   0.143 2.14 0.25        0.48       0.12
```

```
# INTERVENTION EFFECTIVENESS

summary_table$FSW_reduction <-
  1 - (
    summary_table$FSW_prev /
    summary_table$FSW_prev[
      summary_table$Scenario == "Baseline"
    ]
  )

summary_table$MC_reduction <-
  1 - (
    summary_table$MC_prev /
```

```
    summary_table$MC_prev[
      summary_table$Scenario == "Baseline"
    ]
  )

# Rank interventions
summary_table <- summary_table[
  order(summary_table$FSW_reduction,
        decreasing = TRUE),
]

summary_table
```

```
##              Scenario FSW_prev MC_prev   R0   Re FSW_cum_inc MC_cum_inc
## cF8 All_Interventions    0.328   0.143 2.14 0.25        0.48       0.12
## cF5      Condoms_PrEP    0.331   0.147 2.14 0.30        0.52       0.17
## cF6      Condoms_Circ    0.346   0.150 2.14 0.36        0.69       0.20
## cF2      Condoms_Only    0.352   0.158 2.14 0.44        0.77       0.30
## cF7         PrEP_Circ    0.358   0.153 2.14 0.42        0.84       0.24
## cF3         PrEP_Only    0.367   0.162 2.14 0.50        0.95       0.35
## cF4 Circumcision_Only    0.399   0.170 2.14 0.60        1.31       0.45
## cF           Baseline    0.422   0.190 2.14 0.72        1.57       0.69
## cF1   No_Intervention    0.890   0.483 2.14 1.93        6.67       4.17
##     FSW_reduction MC_reduction
## cF8    0.22274882    0.2473684
## cF5    0.21563981    0.2263158
## cF6    0.18009479    0.2105263
## cF2    0.16587678    0.1684211
## cF7    0.15165877    0.1947368
## cF3    0.13033175    0.1473684
## cF4    0.05450237    0.1052632
## cF     0.00000000    0.0000000
## cF1   -1.10900474   -1.5421053
```

```
# RANKED INTERVENTION EFFECTIVENESS

rank_df <- subset(
  summary_table,
  !(Scenario %in%
      c(
        "Baseline",
        "No_Intervention"
      ))
)

rank_long <- data.frame(

  Scenario = rep(rank_df$Scenario, 2),

  Population = c(
    rep("FSW", nrow(rank_df)),
    rep("Male Clients", nrow(rank_df))
  ),
```

```
  Reduction = c(
    rank_df$FSW_reduction,
    rank_df$MC_reduction
  )
)

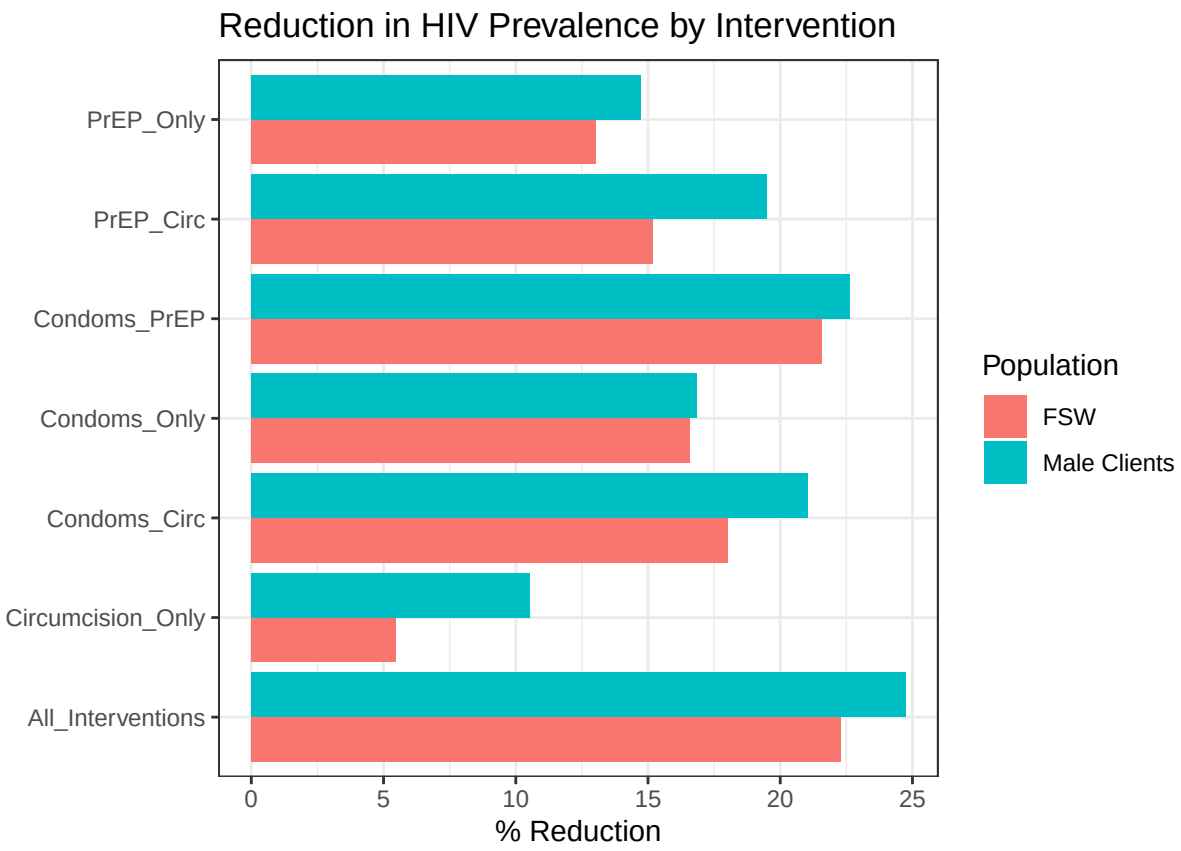
ggplot(
  rank_long,
  aes(
    x = Scenario,
    y = Reduction * 100,
    fill = Population
  )
) +

  geom_col(position = "dodge") +

  coord_flip() +

  theme_bw() +

  labs(
    title = "Reduction in HIV Prevalence by Intervention",
    x = "",
    y = "% Reduction"
  )
```




```r
# Scenario prevalence plots
plot_df <- do.call(
  rbind,

  lapply(names(outputs), function(n){

    out <- outputs[[n]]

    data.frame(

      Time = out$time,

      FSW =
        (out$IF + out$TF) /

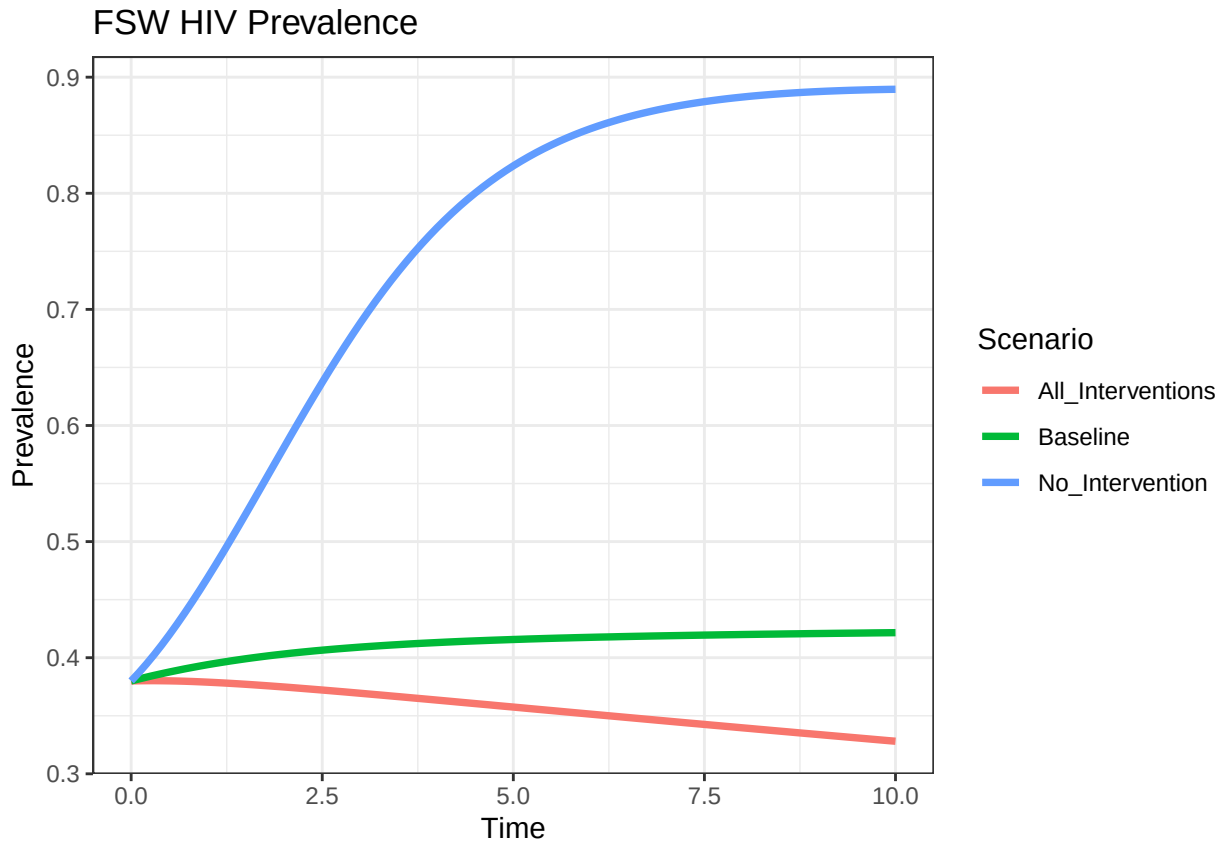
        (out$SF + out$IF + out$TF),

      MC =
        (out$IM + out$TM) /
        (out$SM + out$IM + out$TM),

      Scenario = n
    )
  })
)
```

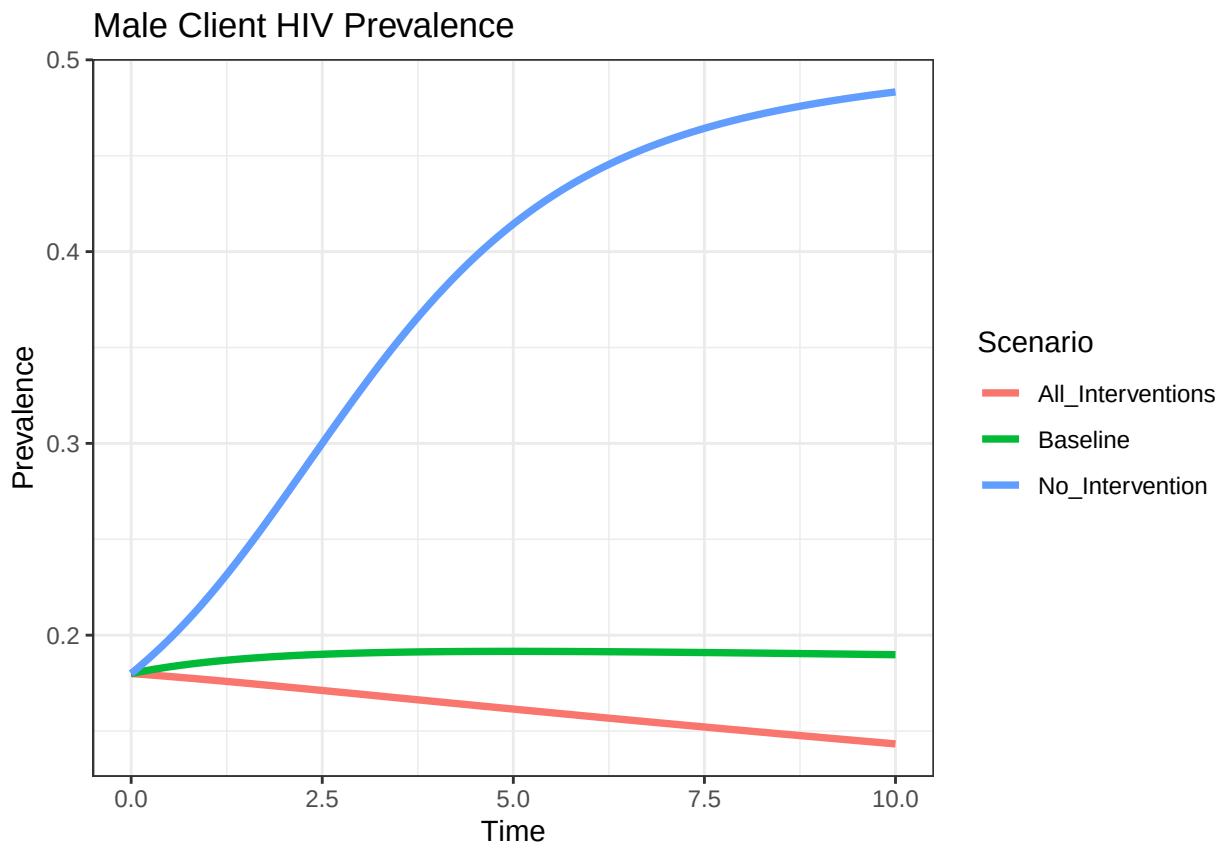
```
plot_df2 <- subset(
  plot_df,
  Scenario %in% c(
    "Baseline",
    "No_Intervention",
    "All_Interventions"
  )
)

ggplot(
  plot_df2,
  aes(Time, FSW, color = Scenario)
) +
  geom_line(linewidth=1.3) +
  theme_bw() +
  labs(
    title="FSW HIV Prevalence",
    y="Prevalence"
  )
```




```
ggplot(
  plot_df2,
  aes(Time, MC, color = Scenario)
) +
  geom_line(linewidth = 1.3) +
```

```
  theme_bw() +
  labs(
    title = "Male Client HIV Prevalence",
    y = "Prevalence"
  )
```




```
# Sensitivity analysis
sens_grid <- expand.grid(

  beta_scale = c(0.5, 0.75, 1, 1.25, 1.5),

  condom_covF = c(0.2, 0.4, 0.6, 0.8),

  PrEP_covF = c(0.0, 0.2, 0.4)

)

sens_results <- data.frame()

for(i in 1:nrow(sens_grid)){

  # copy baseline parameters
  p <- params_calibrated

  # vary transmission probabilities
```

```
  p$betaMF <- p$betaMF * sens_grid$beta_scale[i]
  p$betaFM <- p$betaFM * sens_grid$beta_scale[i]

  # vary interventions
  p$condom_covF <- sens_grid$condom_covF[i]
  p$PrEP_covF   <- sens_grid$PrEP_covF[i]

  # run model
  out <- run_model(p)

  # final prevalence
  FSW_prev <- tail(
    (out$IF + out$TF) /

    (out$SF + out$IF + out$TF),
    1
  )

  MC_prev <- tail(
    (out$IM + out$TM) /
    (out$SM + out$IM + out$TM),
    1
  )

  # cumulative incidence
  FSW_cum <- sum(out$incF)
  MC_cum  <- sum(out$incM)

  sens_results <- rbind(
    sens_results,

    data.frame(
      beta_scale  = sens_grid$beta_scale[i],
      condom_covF = sens_grid$condom_covF[i],
      PrEP_covF   = sens_grid$PrEP_covF[i],

      FSW_prev = FSW_prev,
      MC_prev  = MC_prev,

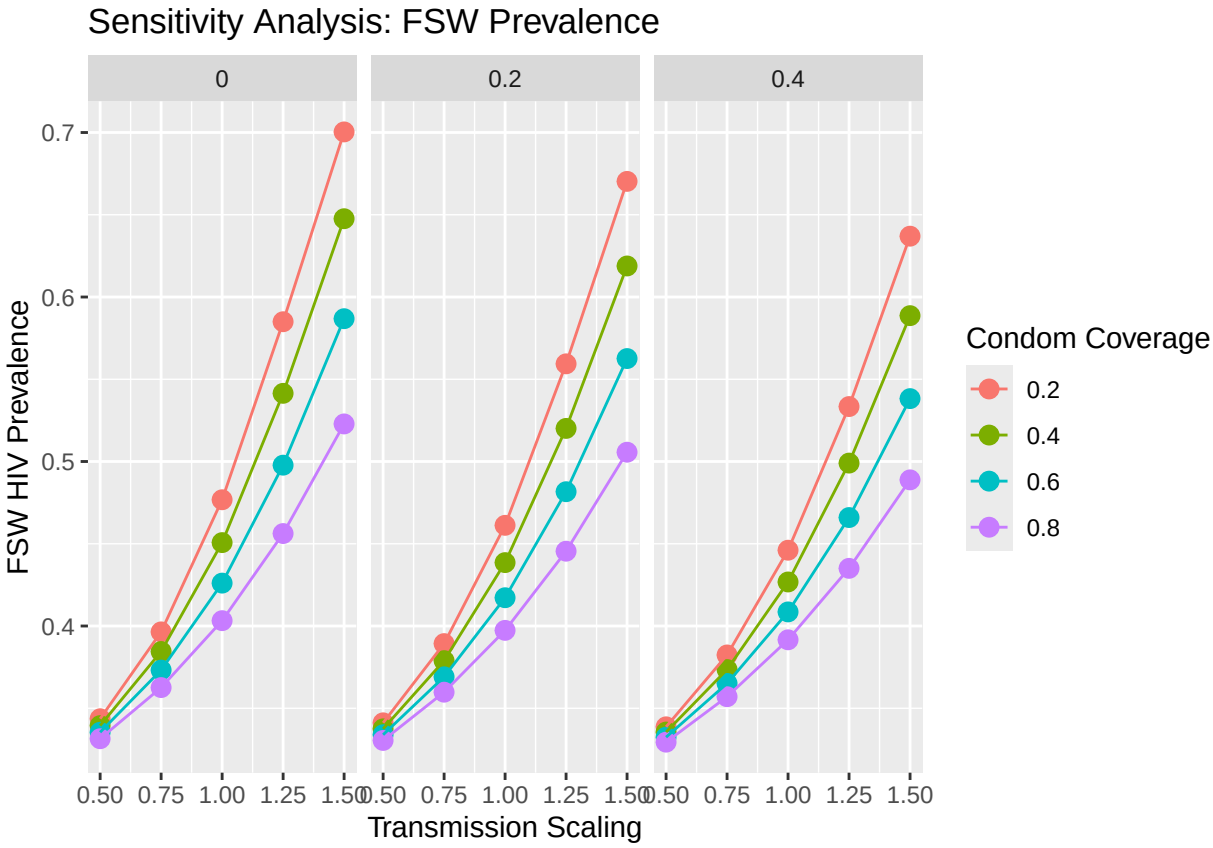
      FSW_cum = FSW_cum,
      MC_cum  = MC_cum
    )
  )
}
# SENSITIVITY PLOTS
# FSW Prevalence
ggplot(
  sens_results,
  aes(beta_scale,
      FSW_prev,
      color = factor(condom_covF))
) +
```

```
  geom_point(size = 3) +

  geom_line() +

  facet_wrap(~PrEP_covF) +

  labs(
    title = "Sensitivity Analysis: FSW Prevalence",
    x = "Transmission Scaling",
    y = "FSW HIV Prevalence",
    color = "Condom Coverage"
  )
```



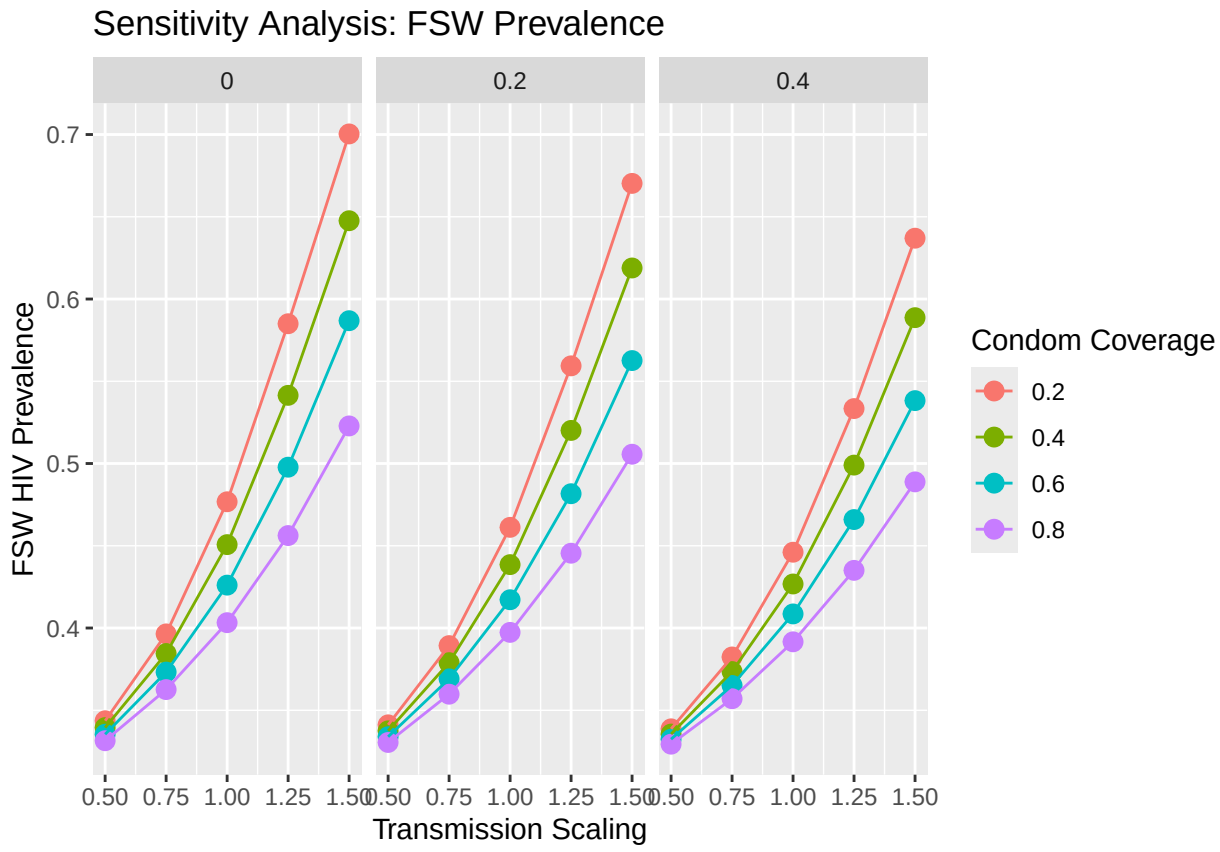

```
  ggplot(
  sens_results,
  aes(beta_scale,
      FSW_prev,
      color = factor(condom_covF))
) +

  geom_point(size = 3) +

  geom_line() +

  facet_wrap(~PrEP_covF) +
```

```
  labs(
    title = "Sensitivity Analysis: FSW Prevalence",
    x = "Transmission Scaling",
    y = "FSW HIV Prevalence",
    color = "Condom Coverage"
  )
```




```
# Male client prevalence
  ggplot(
  sens_results,
  aes(beta_scale,
      MC_prev,
      color = factor(condom_covF))
) +

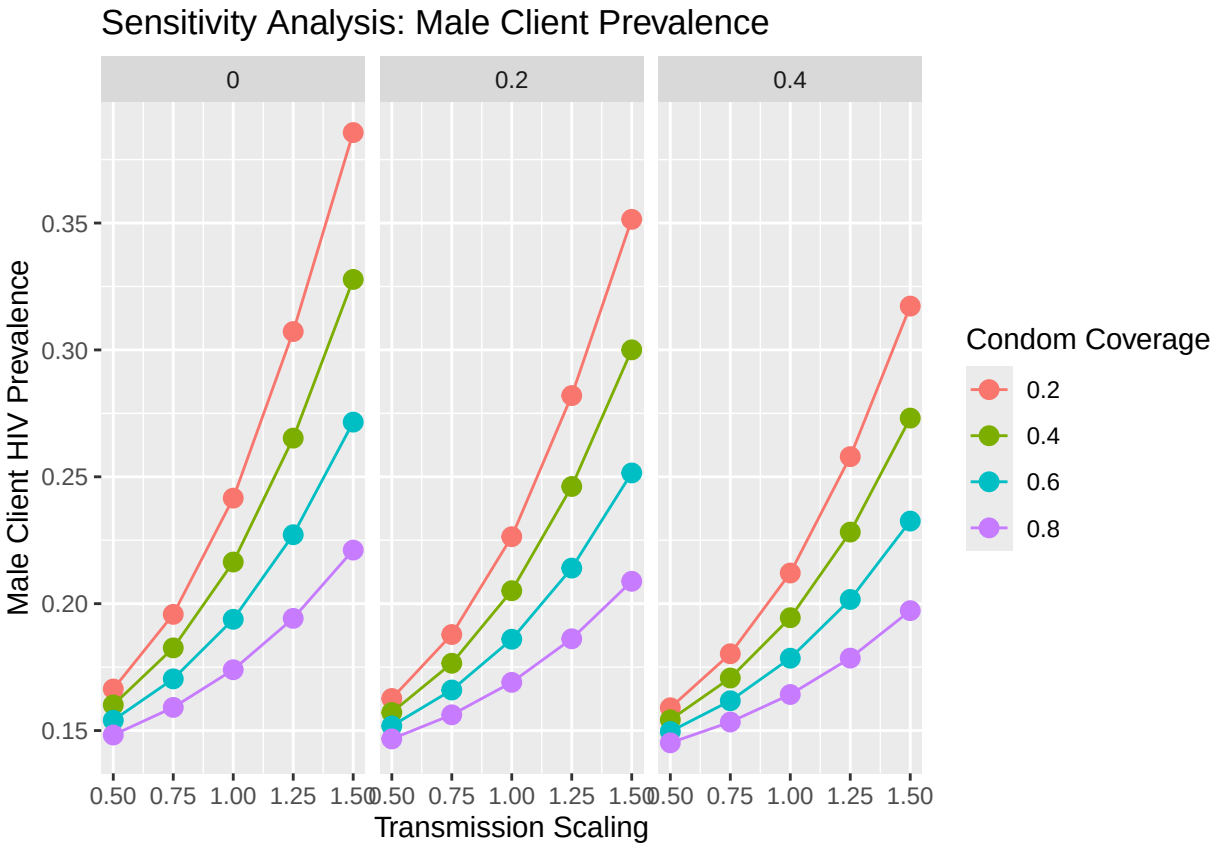
  geom_point(size = 3) +

  geom_line() +

  facet_wrap(~PrEP_covF) +

  labs(
    title = "Sensitivity Analysis: Male Client Prevalence",
    x = "Transmission Scaling",
    y = "Male Client HIV Prevalence",
```

```
    color = "Condom Coverage"
  )
```




```
# Heat maps
coverage_grid <- expand.grid(

  condom_covF = seq(0, 0.9, by = 0.1),

  PrEP_covF = seq(0, 0.8, by = 0.1)

)

heatmap_results <- data.frame()

# Heat map analysis
for(i in 1:nrow(coverage_grid)){

  p <- params_calibrated

  p$condom_covF <- coverage_grid$condom_covF[i]
  p$PrEP_covF   <- coverage_grid$PrEP_covF[i]

  out <- run_model(p)

  # cumulative incidence
```

```
  FSW_cum <- sum(out$incF)
  MC_cum  <- sum(out$incM)

  # prevalence
  FSW_prev <- tail(
    (out$IF + out$TF) /

    (out$SF + out$IF + out$TF),
    1
  )

  MC_prev <- tail(
    (out$IM + out$TM) /
    (out$SM + out$IM + out$TM),
    1
  )

  heatmap_results <- rbind(
    heatmap_results,

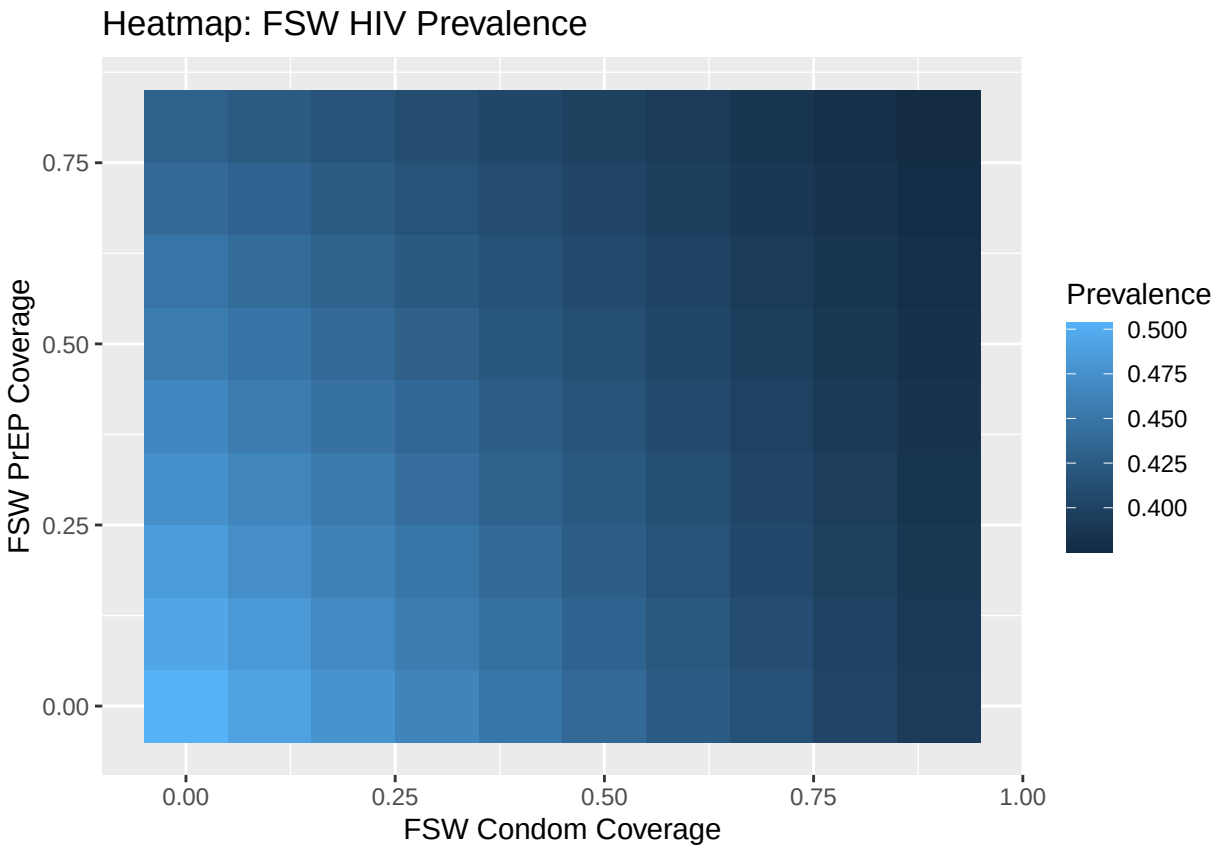
    data.frame(
      condom_covF = p$condom_covF,
      PrEP_covF   = p$PrEP_covF,

      FSW_cum = FSW_cum,
      MC_cum  = MC_cum,

      FSW_prev = FSW_prev,
      MC_prev  = MC_prev
    )
  )
}
# Heat map FSW prevalence
ggplot(
  heatmap_results,
  aes(
    x = condom_covF,
    y = PrEP_covF,
    fill = FSW_prev
  )
) +

  geom_tile() +

  labs(
    title = "Heatmap: FSW HIV Prevalence",
    x = "FSW Condom Coverage",
    y = "FSW PrEP Coverage",
    fill = "Prevalence"
  )
```


Heatmap: FSW HIV Prevalence
FSW PrEP Coverage
0.00
0.25
0.50
0.75
FSW Condom Coverage
0.00
0.25
0.50
0.75
1.00
Prevalence
0.500
0.475
0.450
0.425
0.400